LECTURE 19

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$$\underbrace{\int_1^\infty \frac{1}{x^p}\,dx}_{p\text{ test}} = \lim_{b\to\infty}\int_1^b \frac{1}{x^p}\,dx = \lim_{b\to\infty} \left.\frac{x^{-p+1}}{1-p}\right|_1^b = \lim_{b\to\infty}\frac{b^{-p+1}}{1-p} - \frac{1}{1-p}$$

$$\Updownarrow \qquad = \frac{1}{1-p} \qquad p>1 \quad \text{cgt.}$$

$$\infty \qquad p\le 1 \quad \text{div.}$$

$$\sum_{n=1}^{\infty}\frac{1}{n^p} \qquad \begin{array}{l} p>1 \text{ cgt} \\ p\le 1 \text{ div} \end{array}$$

General form: $\displaystyle\int_a^\infty \frac{1}{x^p}\,dx$ $\quad p>1$ cgt, $\quad p\le 1$ div.

Limit comp. Test. for improper integral of IInd kind:

If $\displaystyle\lim_{x\to\infty}\frac{f(x)}{g(x)} \ne 0$ finite then $\displaystyle\int_a^\infty f(x)\,dx$ & $\displaystyle\int_a^\infty g(x)\,dx$ behave lly.

Eg 1) $\displaystyle\int_1^\infty \frac{dx}{x\sqrt{1+x^2}}\,dx.$ $\qquad f(x) = \dfrac{1}{\underset{\downarrow\, 1}{x}\,\underset{\downarrow\, 1}{\sqrt{1+x^2}}}$ $\qquad \underbrace{g(x) = \dfrac{1}{x^2}}$

$$\lim_{x\to\infty}\frac{f(x)}{g(x)} = \lim_{x\to\infty}\frac{x^2}{x\sqrt{1+x^2}} \qquad \int_1^\infty \frac{1}{x^2}\,dx = \text{cgt.}$$

$$= \lim_{x\to\infty}\frac{1}{\sqrt{\frac{1}{x^2}+1}} = 1 \ne 0 \text{ finite}$$

2) $\displaystyle\int_2^\infty \frac{1}{\underbrace{\sqrt{x^2-x-1}}}\,dx =$ div. $\qquad g(x) = \dfrac{1}{x} \qquad \underbrace{\displaystyle\int_2^\infty \frac{1}{x}\,dx}$ - div

$$\lim_{x\to\infty}\frac{x}{\sqrt{x^2-x-1}} = 1. \ne 0 \text{ finite}$$

$f(x) =$ $[1, \infty]$

$= 1.$

3) $\displaystyle\int_1^\infty \frac{1}{x^2(1+e^x)}$ $\quad f(x) = \dfrac{1}{x^2(1+e^x)}$ $\quad [1, \infty]$

$\downarrow$ cgt $\quad |f(x)| \le \displaystyle\int_1^\infty \frac{1}{x^2}$ cgt. $\quad \dfrac{1}{1+(+ve)} \le 1$

Comparison test:

4) $\displaystyle\int_\pi^\infty \frac{\sin x}{x^2}\,dx$ cgt. $\quad |f(x)| = \left|\dfrac{\sin x}{x^2}\right| \le \dfrac{1}{x^2}$

$\displaystyle\int_\pi^\infty \frac{1}{x^2}$ cgt.

H.W. $\displaystyle\int_0^\infty e^{-x}\frac{\sin x}{x}\,dx$

$$\int_0^1 \frac{1}{x^p}\,dx = \lim_{\epsilon\to 0}\int_\epsilon^1 \frac{1}{x^p}\,dx = \lim_{\epsilon\to 0}\frac{\epsilon^{-p+1}}{1-p} - \frac{1}{1-p}$$

$= \infty$ div $\quad p \ge 1$

$= \dfrac{1}{p-1}$ cgt $\quad p < 1$

General form: $\displaystyle\int_a^b \frac{dx}{(x-a)^p}$ $\quad p<1$ cgt, $\quad p \ge 1$ div.

Limit. Comp. Test for Improper integrals of IInd kind

If $\displaystyle\lim_{x\to a}\frac{f(x)}{g(x)} \ne 0$ finite then $\displaystyle\int_a^b f(x)\,dx$ & $\displaystyle\int_a^b g(x)\,dx$ behave ||ly.

Eg. 1) $\displaystyle\int_0^1 \frac{1}{x^3(1+x^2)}\,dx$ div. $\quad$ Note: choose bad part as $g(x)$

$g(x) = \dfrac{1}{x^3}$

$$\lim_{x\to 0}\frac{f(x)}{g(x)} = \lim_{x\to 0}\frac{1}{1+x^2} = 1 \ne 0 \text{ finite}$$

$\displaystyle\int_0^1 \frac{1}{x^3}\,dx$ div.

2) $\displaystyle\int_0^{\pi/2} \frac{\cos x}{x^2}\,dx$ div. $\quad g(x) = \dfrac{1}{x^2}$ $\quad \displaystyle\int_0^{\pi/2} g(x)\,dx = \int_0^{\pi/2}\frac{1}{x^2}\,dx$ div.

2) $\int_0^{\cdot} \frac{\cdot}{x^2}\, dx$ div. $g(x) = \frac{1}{x^2}$ $\int_0 g(x)\, dx = \int_0 \frac{1}{x^2}\, dx$ div.

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = \lim_{x \to 0} \frac{\cos x \sqrt{x^2}}{1/x^2} = 1 \neq 0 \text{ finite}$$

3) $\int_0^1 \frac{1}{x^{1/3}(1+x^2)}$